Contribution Title

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Abstract. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.
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Keywords: First keyword · Second keyword · Another keyword.

1 Introduction

Prostate-specific membrane antigen (PSMA) PET/CT has become a central imaging modality for staging, treatment planning, radiopharmaceutical therapy, and post-therapy response assessment in prostate cancer [6]. In these settings, patients often undergo multiple scans over time, comprising a baseline scan before therapy and one or more follow-up scans afterwards; accurate baseline-to-follow-up registration can support comparison of disease burden, localisation of uptake changes, assessment of lesion progression or response, dose accumulation, and quantitative longitudinal analysis [12]. The challenge organisers identify several factors that make establishing this correspondence by deformable image registration difficult in routine whole-body data [12]: baseline and follow-up scans differ substantially in field of view, patient positioning, respiratory state, bladder and bowel filling, and body weight; CT and PET encode complementary but appearance-dissimilar information; and the registration must preserve the quantitative PET signal used for response assessment while remaining physically plausible structures such as the skeleton. The Learn2Reg 2026 PSMAReg challenge [9, 8, 10] formalises this problem as the estimation of a dense displacement field that warps each follow-up (moving) scan to its baseline (fixed) scan, evaluated jointly on organ overlap, boundary distance, deformation regularity, and preservation of PET-derived tumour biomarkers.

Deformable registration has traditionally been posed as an iterative optimisation over a regularised similarity objective, with widely used instances including the diffeomorphic Demons [22], SyN [1], and B-spline [19] schemes. While accurate, these methods are slow at whole-body resolution. Learning-based methods instead amortise the optimisation over a training set into the network weights, replacing per-pair iterative optimisation with a single forward pass at inference. VoxelMorph established the unsupervised paradigm [3], and subsequent work introduced diffeomorphic parameterisations through stationary velocity fields [5], transformer backbones such as TransMorph [4], and multi-resolution architectures such as LapIRN [16]. Among the latter, LapIRN uses a Laplacian pyramid of coarse-to-fine sub-networks to recover large deformations while keeping the transformation smooth and invertible, and has been a strong performer across previous Learn2Reg editions [9, 8]. A recurring limitation of purely intensity-driven losses is that they permit non-physical folding. Anatomy-aware regularisation that penalises non-rigid deformation, e.g. of skeletal structures [14, 18, 20, 15], is therefore particularly relevant for whole-body PET/CT, where the skeleton is a frequent site of PSMA-avid disease.

In this work, we adapt LapIRN to longitudinal whole-body PSMA PET/CT for the PSMAReg task.

Our contributions are:

- an adaptation of the diffeomorphic, coarse-to-fine LapIRN architecture to longitudinal whole-body PSMA PET/CT, combining an affine pre-registration with a three-level pyramid and test-time instance optimisation;
- a set of biomarker-preserving losses that constrain the PET-derived MTV and TLG under the estimated deformation, including a Jacobian-based surrogate that supplies dense gradients throughout the lesion interior;
- a per-structure rigid-fit bone-rigidity penalty
- a gradient-level analysis of the objectives showing the accuracy-versus-PET-preservation trade-off;
- a score-driven checkpoint and submission selection procedure that mirrors the challenge ranking. [TODO: finalise once the model is fixed]

The proposed registration pipeline with affine pre-registration, LapIRN-based registration, losses and instance optimization (IO) is shown in Fig. 1.

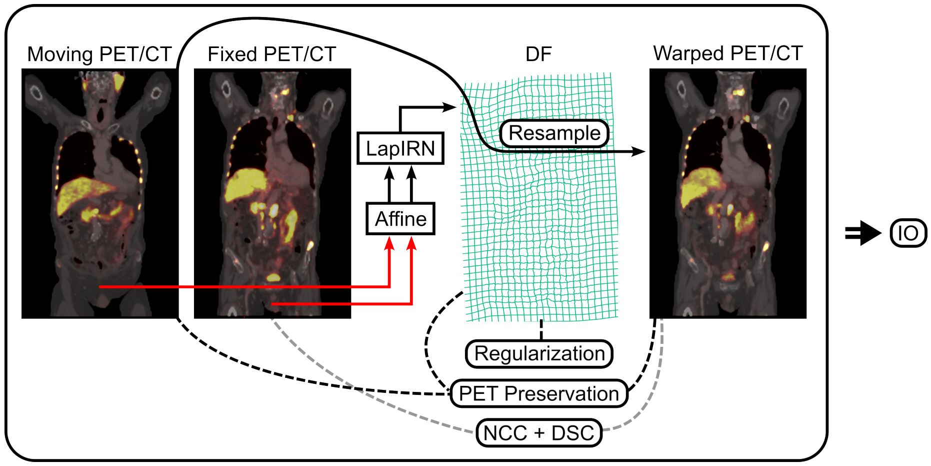

Fig. 1. Overview of the proposed registration pipeline. The moving (follow-up) and fixed (baseline) PET/CT are first coarsely aligned by an affine pre-registration (red), and the affinely aligned pair is passed to the coarse-to-fine LapIRN network, which predicts a dense displacement field (DF, green grid) used to resample the moving scan into the fixed space (warped PET/CT). Training is driven by three groups of objectives: deformation regularisation on the DF, comprising a global smoothness and folding penalty and a per-structure bone-rigidity term that constrains the DF inside skeletal labels and thereby also acts as an anatomical surrogate for biomarker preservation, since a large fraction of the lesion volume is intraosseous; PET-biomarker preservation terms (MTV, TLG and Jacobian inside the tumour mask) comparing the moving and warped lesions (black dashed); and image- and label-based similarity, NCC and DSC, between the fixed and warped scans (grey dashed). At test time, the same objectives are descended per pair by instance optimisation (IO).

2 Methods

2.1 Data and Preprocessing

We used the PSMAReg dataset of the Learn2Reg 2026 challenge [12], a longitudinal whole-body PSMA PET/CT cohort adapted from the manually annotated autoPET data of [10]. Each paired subject provides a baseline scan and one or more follow-up scans, each with a CT and a PSMA PET channel; the follow-up is registered to the baseline, i.e. the baseline acts as the fixed and the follow-up as the moving image.

The challenge training split contains 146 paired and 212 unpaired subjects; the unpaired single-time-point cases were not used in this work. We divided the paired subjects patient-wise into 117 training and 29 internal validation subjects. Registration pairs are formed within a subject and always point from the temporally later to the earlier scan. For training we used all such ordered time-point combinations, including follow-up-to-follow-up pairs, which yields 239 pairs from the 117 subjects and acts as a mild augmentation of the 146 baseline-anchored pairs. For our internal validation subjects we kept only baseline-anchored pairs, i.e. every follow-up registered to its baseline, matching the pairing used at evaluation time and giving 39 pairs from the 29 subjects. The challenge validation split of 20 paired subjects, which is the split scored on the public validation leaderboard, served as our test set; the organisers' own held-out test set ($\sim$200 pairs of private JHU data) is scored only by them.

For the training split we used the labels released by the organisers: CT organ labels generated automatically with TotalSegmentator [23] and PET lesion labels manually annotated by radiologists. Since no labels are released for the challenge validation split, we generated our own for our test set with TotalSegmentator for the CT organs and the winning autoPET III submission [17] for the PET lesions.

All images are provided already resampled to an isotropic in-plane spacing of $2.7344 \times 2.7344 \times 3.27$ mm and cropped to a fixed size of $192 \times 192 \times 288$ voxels, so no further geometric preprocessing was applied. We normalised CT by clipping to a fixed window of $[-1000, 1500]$ HU and PET by clipping the SUV to $[0, 20]$, each rescaled to $[0, 1]$.

2.2 Registration Pipeline

The pipeline consists of an affine pre-registration followed by a deformable LapIRN stage and, at test time, instance optimisation (Fig. 1). Throughout, I denotes the moving PET image, M its lesion mask, φ_{aff} the affine pre-registration, φ_θ the displacement field predicted by the network, and

$$\varphi = \varphi_{\mathrm{aff}} \circ \varphi_\theta \tag{1}$$

the total transformation mapping the moving into the fixed image; W denotes the lesion region in the fixed frame, i.e. the image of M under φ.

Affine pre-registration. Baseline and follow-up scans of the same subject differ in field of view, table position and patient posture by far more than a locally regularised network can absorb, so we first remove the global component with a separate affine stage. For each pair we estimate an affine transform with ANTs [2] on the CT channel alone, at half resolution and with a Mattes mutual-information metric, after masking both volumes to the body and windowing them to $[-1000, 1000]$ HU. The result is converted into a dense voxel displacement field, cached per pair, and applied to the moving scan before the network sees it; augmentations are applied consistently to the cached field. Note that an affine transform is in general not volume-preserving, which is why all biomarker terms of Sec. 2.3 are evaluated on the composed transform (1) rather than on φ_θ alone.

Deformable registration. For the deformable stage we use LapIRN [16], a Laplacian-pyramid network whose three sub-networks operate at a quarter, a half and the full resolution of the $192 \times 192 \times 288$ volumes. Each level predicts a stationary velocity field that is added to the upsampled velocity of the coarser level; the sum is exponentiated by scaling and squaring to give a diffeomorphic φ_θ, so large deformations are recovered coarse-to-fine while the transformation stays smooth and invertible. We adapted the architecture to the multi-modal, whole-body setting: each sub-network takes four input channels, the CT and PSMA PET of the affinely aligned moving scan and of the fixed scan, and uses 24 initial feature channels with five residual blocks per level. Image similarity is computed on CT only, since PET uptake is not conserved between time points and would otherwise drive the deformation towards matching disease rather than anatomy; the PET channel enters the network as input and the biomarker terms of Sec. 2.3 instead. The levels are trained in sequence, each initialised from the previous one, which is kept frozen for the first ten epochs and fine-tuned jointly afterwards.

2.3 Loss Functions

The full objective combines image similarity, weak label supervision, the PET-biomarker terms, the bone-rigidity penalty and two deformation regularisers:

$$\begin{aligned}\mathcal{L} = \lambda_{\mathrm{NCC}}\,\mathcal{L}_{\mathrm{NCC}} + \lambda_{\mathrm{DSC}}\,\mathcal{L}_{\mathrm{DSC}} + \lambda_{\mathrm{MTV}}\,\mathcal{L}_{\mathrm{MTV}} + \lambda_{\mathrm{MTV\text{-}J}}\,\mathcal{L}_{\mathrm{MTV\text{-}J}} \\ + \lambda_{\mathrm{jac}}\,\mathcal{L}_{\mathrm{jac}} + \lambda_{\mathrm{TLG}}\,\mathcal{L}_{\mathrm{TLG}} + \lambda_{\mathrm{rigid}}\,\mathcal{L}_{\mathrm{rigid}} + \lambda_{\mathrm{smooth}}\,\mathcal{L}_{\mathrm{smooth}} + \lambda_{\mathrm{NDV}}\,\mathcal{L}_{\mathrm{NDV}}.\end{aligned} \tag{2}$$

The weights are given in Table **??** [TODO: add weight table once the final run is fixed; current values: $\lambda_{\mathrm{NCC}} = 5$, $\lambda_{\mathrm{DSC}} = 3/4/5$ for levels 1/2/3, $\lambda_{\mathrm{MTV}} = 20$, $\lambda_{\mathrm{MTV\text{-}J}} = 0.5$, $\lambda_{\mathrm{jac}} = 5$, $\lambda_{\mathrm{TLG}} = 5$, $\lambda_{\mathrm{rigid}} = 0.2$, $\lambda_{\mathrm{smooth}} = 5$, $\lambda_{\mathrm{NDV}} = 10^4$]. All terms are evaluated on the composed transformation (1). Only the full-resolution level is trained with the complete objective. The two coarser levels use the similarity, label and regularisation terms alone, i.e. $\lambda_{\mathrm{MTV}} = \lambda_{\mathrm{MTV\text{-}J}} = \lambda_{\mathrm{jac}} = \lambda_{\mathrm{TLG}} = \lambda_{\mathrm{rigid}} = 0$ at a quarter and half resolution, because the lesion and bone masks they would be evaluated on are only a few voxels wide there: after downsampling, thin ribs and small lesions are largely partial-volume, so both the volume ratios and the per-structure rigid fits measure interpolation rather than anatomy.

Image similarity. Image similarity is measured by the local normalised cross correlation, evaluated as a multi-resolution sum over as many scales as the level has resolutions, i.e. one scale at the coarsest and three at the full-resolution level. The window is 5^3 voxels at the quarter-resolution level and 7^3 at the half- and full-resolution levels, so that the coarsest level, where one voxel already covers roughly a centimetre, is not asked to match structure over a neighbourhood far larger than the deformations it estimates. It is computed on the CT channel only ($\lambda_{\mathrm{PET}} = 0$): PET intensity is not conserved between the two time points, as lesions appear, resolve or change uptake under therapy, so a PET similarity term would reward matching disease rather than anatomy.

Weak label supervision (DSC). We used a standard DSC loss for weak label supervision. Since the challenge scores PET organ overlap but provides no PET organ annotations for training, we measured each of the 117 TotalSegmentator organs' mean SUV relative to skeletal muscle across all sessions. As 90 organs fall below twice the muscle background and are thus indistinguishable in PET, we upweighted only the 10 exceeding 4 times the background (kidneys, gallbladder, spleen, liver, duodenum, bladder, portal vein, prostate, pancreas) by a factor of 2 in the per-class Dice loss. This relies on PET and CT being well aligned within a session, so that CT-derived overlap on these organs approximates the PET organ overlap being scored.

Preservation of PET quantification. The scored MTV bias only constrains the *net* lesion volume, so we penalize it directly while supplying dense gradients through a Jacobian-based surrogate. All terms are evaluated on the total transformation $\varphi = \varphi_{\text{aff}} \circ \varphi_\theta$, i.e. the affine pre-registration composed with the predicted displacement, against the original moving lesion mask M, since the affine itself is generally not volume-preserving. First, we minimize the squared relative volume error

$$\mathcal{L}_{\text{MTV}} = \left(\frac{|M \circ \varphi| - |M|}{|M|} \right)^2, \tag{3}$$

where $M \circ \varphi$ denotes the linearly resampled mask. This matches the scored metric exactly, but its gradient is confined to the lesion's sub-voxel boundary shell. We therefore add a complementary term based on the change-of-variables identity: summing $\det J_\varphi$ over the warped lesion region recovers the original lesion volume, so the mean determinant over the warped lesion equals the ratio of original to warped volume. Driving this mean to one,

$$\mathcal{L}_{\text{MTV-J}} = \left(\frac{1}{|W|} \sum_{x \in W} \det J_\varphi(x) - 1 \right)^2, \tag{4}$$

where W denotes the warped lesion region, therefore pins the same net volume ratio as (3) while providing gradients throughout the lesion interior; unlike a per-voxel constraint, it still permits internal redistribution of volume. Note that masking the determinant with the moving-frame lesion, as is tempting, samples $\det J_\varphi$ at locations unrelated to the lesion once the affine is composed in; empirically, this decorrelated the surrogate from the scored MTV (Fig. 2). Finally, the per-voxel penalty

$$\mathcal{L}_{\text{jac}} = \frac{1}{|W|} \sum_{x \in W} \left(\det J_\varphi(x) - 1 \right)^2 \tag{5}$$

acts as a stricter local regulariser, discouraging compensating compression and expansion within the lesion.

FIGURE 2 – Schematic, two panels. (a) masking $\det J_\varphi$ *with the moving-frame lesion mask: once the affine is composed in, the sampled locations are unrelated to the lesion. (b) masking with the warped lesion region* W*: the sampled locations coincide with the lesion. This is the subtle point of Eq. (2) and a diagram makes it instant. High value per column inch.*

Fig. 2. Moving-frame versus warped-frame masking of the Jacobian determinant. [TODO: caption]

The challenge additionally scores the total lesion glycolysis (TLG), the PET-activity–weighted counterpart of the MTV, which we penalize analogously as the relative change of intensity mass under the warp:

$$\mathcal{L}_{\mathrm{TLG}} = \frac{\left|\sum_x (I \circ \varphi)(x)\,(M \circ \varphi)(x) - \sum_y I(y)\,M(y)\right|}{\sum_y I(y)\,M(y)}, \tag{6}$$

where I denotes the moving PET image. Since both the image and the mask are resampled with the same transformation, this term is sensitive not only to volume change but also to intensity blurring at the lesion boundary, and thus complements the purely geometric terms (3)–(5).

The same terms (3)–(6) are used during training and descended at test time by our instance optimisation (Sec. 2.4), so the network is trained on the objective it is later refined under.

Bone rigidity. In 146 patients with paired baseline and follow-up scans, a median of 46.0% (IQR 3.1 to 72.3%) of total lesion volume at baseline lay within skeletal structures, rising to 63.8% (IQR 12.1 to 77.4%) at follow-up. Since a substantial fraction of lesion volume is therefore intraosseous, a rigidity penalty on a bone mask is an anatomically grounded surrogate for the biomarker terms: enforcing rigid bone motion is both physiologically correct and, where lesions sit in bone, directly prevents the volume change MTV measures.

The established way of promoting local rigidity is a variational penalty on the Jacobian J_φ of the transformation. A deformation is locally rigid where J_φ is orthogonal, i.e. $J_\varphi^T J_\varphi = \mathrm{Id}$, so deviations from orthogonality are penalised inside the rigid regions [14, 18]. Building on this, [20, 15] additionally enforce local affineness, by penalising deviations of the second-order derivatives $\partial^2 u$ from zero, and orientation preservation, by penalising deviations of $\det J_\varphi$ from unity; combining the three conditions preserves rigidity better than any of them alone. We adopted this three-condition penalty first. All of its terms are evaluated by finite differences at every voxel and only then masked, so that the mask selects which voxels are scored but not which are read. The derivative stencil therefore also reads non-bone voxels, and with thin cortical bone and ribs this dominates: the term ends up measuring the bone-to-soft-tissue displacement discontinuity that any correct registration must exhibit. We quantify this failure mode in Sec. 3.2. Eroding the mask would remove the effect, but in the present dataset many bone masks are only a few voxels wide.

We therefore replaced this loss with a per-structure rigid-fit residual: for each label, we obtain the optimal rotation and translation from the predicted displacement in closed form via Kabsch alignment [11], and penalise the mean squared residual of that fit. No stencil crosses the mask, and each label is fitted independently, so articulation is free. The penalty constrains only the shape of the deformation within each structure and is invariant to its placement, leaving alignment to NCC and Dice, unlike derivative based penalties [20, 15] and unlike [7], who regress the field onto a per bone rigid field precomputed by Dice pre registration, requiring segmentations in both images and inheriting pre registration error as a training target. The penalty is applied to the 61 skeletal TotalSegmentator structures. For each structure separately, the voxel positions and their displaced counterparts under the composed field form two point sets, and the rigid transform relating them best in the least-squares sense follows from the singular value decomposition of their centred covariance, with a sign correction that forbids a reflection [21] and so takes over the role of the orientation-preservation condition above. The loss is the mean squared distance of the displaced voxels from that fit, averaged first within and then across structures, so a rib counts as much as the pelvis; structures below 50 voxels are skipped, as the fit is ill-posed for a handful of points. Because the fitted transform is itself the minimiser of that residual, the derivative of the loss with respect to it vanishes, so it can be detached and the gradient taken through the displacement alone without approximation, avoiding a backward pass through the SVD.

Deformation regularisation. To regularise the displacement field we used a diffusion regulariser, the mean squared spatial gradient of the displacement,

$$\mathcal{L}_{\text{smooth}} = \frac{1}{3} \sum_{d \in \{x,y,z\}} \frac{1}{|\Omega|} \sum_{p \in \Omega} \left\| \partial_d u(p) \right\|^2, \tag{7}$$

together with a term penalising non-diffeomorphic voxels. Rather than the usual penalty on negative Jacobian determinants of a single finite-difference stencil, we differentiably reproduce the scored metric itself: the challenge computes the non-diffeomorphic volume by decomposing each voxel into six tetrahedra and accumulating the negative part of their determinants over the body mask [13]. Writing $\det J_\varphi^{(k)}$ for the determinant of the k-th tetrahedral scheme, the loss

$$\mathcal{L}_{\text{NDV}} = \frac{100}{12|\Omega|} \sum_{p \in \Omega} \sum_{k=1}^{6} \max\bigl(-\det J_\varphi^{(k)}(p),\, 0\bigr) \tag{8}$$

equals the scored percentage of non-diffeomorphic volume and places gradients on exactly the folded voxels, so that minimising it minimises the reported %NDV directly.

Losses considered and rejected. We deliberately do not add a differentiable surface-distance term (e.g. the distance-transform Hausdorff surrogate of Karimi and Salcudean [TODO: citation]) to either the training objective or the instance optimisation: a per-label analysis of the validation predictions shows that most labels already lie at the discretisation floor, while the mean HD95 is driven by a tail that a boundary loss cannot address. The supporting analysis is given in Sec. 3.2.

2.4 Instance Optimisation

At test time we refine each pair individually. Keeping the predicted field fixed, we optimise a residual stationary velocity field, initialised to zero and exponentiated by scaling and squaring, on top of it, so the refined transformation stays diffeomorphic by construction and the network is only ever corrected, never overwritten. The objective is the same one the network was trained under, only reweighted (Table **??**). We descend it for 90 Adam steps with a learning rate of 10^{-2} and keep the best rather than the last iterate, scoring each step with the hard, nearest-neighbour-counted DSC, HD05, MTV and TLG instead of the bilinear proxies the optimiser differentiates, so the selected step is the one the scorer would prefer.

2.5 Model Selection

Model selection enters our pipeline twice: once *within* a run, to decide which checkpoint to keep, and once *across* runs, to decide which variant to submit in the container. Both use the same three-component quality notion, and differ only in how each component is scored.

Both stages combine registration accuracy (CT DSC and HD95), PET-biomarker preservation (absolute MTV and TLG bias) and deformation regularity (%NDV after the official 0.005% practical-equivalence adjustment) as a weighted geometric mean with exponents 0.4, 0.4 and 0.2, mirroring the challenge's final score

$$S = 100 \cdot A^{0.4}\, B^{0.4}\, R^{0.2}, \tag{9}$$

where the accuracy term A is the geometric mean of the DSC and HD95 scores, the biomarker term B that of the MTV and TLG scores, and R the %NDV score.

Checkpoint selection. Registration accuracy and tumour preservation do not peak together: in level-3 training, validation Dice improves until it plateaus, whereas the tumour metrics reach their optimum roughly halfway through and

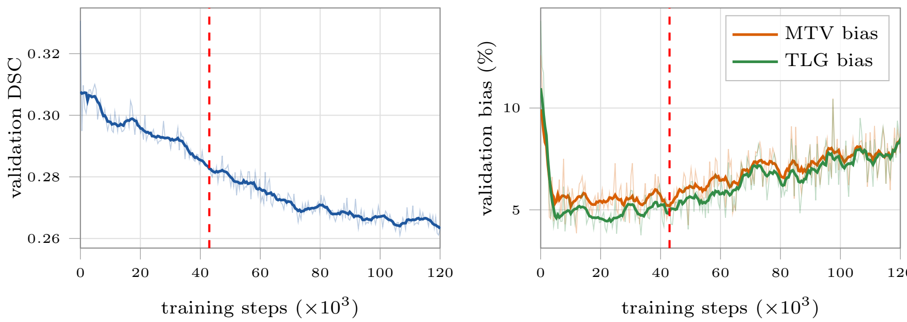

Fig. 3. Accuracy and biomarker preservation do not peak at the same checkpoint. Validation CT Dice (left) rises to a plateau, while the MTV and TLG bias (right) reach their optimum roughly halfway through level-3 training and degrade thereafter. The dashed line marks the checkpoint selected by Eq. (9). Faint lines are the per-round values, solid lines a centred rolling mean.

degrade thereafter (Fig. 3). Selecting on Dice alone therefore yields a checkpoint well past the tumour optimum, so we select on Eq. (9) instead. The official per-metric scores are ranks within a cohort of competing methods, which does not exist during a single training run. We therefore replace each rank score with a logistic quality centred on the metric's median and scaled by its robust spread (interquartile range divided by 1.349), both measured over the validation rounds of a completed run.

Submission selection. Choosing between trained variants does provide a cohort, so here we evaluate Eq. (9) exactly as the challenge does: each of the five metrics is ranked separately across the K variants, with ties taking the worst rank they span, and rank r is turned into the score $(K - r + 1)/K$. The variant with the highest S is the one we submit. To check that this choice is not an artefact of a few cases, we additionally compare the selected variant against every other variant per metric with a paired Wilcoxon signed-rank test over the validation cases, Holm-corrected within each metric. As the variant is selected on the same cases, these tests are exploratory rather than confirmatory; the official validation leaderboard serves as the held-out check.

2.6 Baselines

We compare against two references and three competing methods, all evaluated on the same pairs and under the identical protocol of Sec. 3.1. The two references bound the range in which a deformable method has to be useful: the unregistered pair, i.e. the identity transform, and our affine pre-registration alone, which is also the initialisation every method below starts from.

As a classical iterative baseline we ran NiftyReg [TODO: cite NiftyReg], using `reg_aladin` for the affine stage followed by `reg_f3d` initialised with it, so that the resulting control point grid encodes the total transform; all parameters were left at their defaults. As a learning-free optimisation baseline we ran Convex-Adam [?], in the form provided by the challenge organisers as the official baseline of the task, so that our numbers are directly comparable to the reference entry on the leaderboard. Finally, we compare against vanilla LapIRN, i.e. the same architecture and training schedule as our model but with the original objective only, without the DSC, NDV loss, PET-biomarker terms and the bone-rigidity penalty, which isolates the contribution of the proposed losses from that of the backbone.

2.7 Implementation Details

3 Results

3.1 Evaluation Protocol

The challenge does not release labels for the validation cases. Unless stated otherwise, the validation numbers we report are therefore computed against surrogate labels we generated ourselves: CT organ labels with TotalSegmentator [?] and PET tumour labels with the winning autoPET-III submission [?]. These surrogate labels underlie every validation metric in this work, including the checkpoint and submission selection of Sec. 2.5. Being predictions rather than the challenge's reference annotations, they carry their own error, so their absolute values are not directly comparable to the official validation leaderboard; we use them to compare methods against each other under an identical protocol. Where we report official validation-leaderboard results, these are computed by the organisers against their reference annotations and are marked as such. Since the leaderboard covers the same cases with the correct labels, we use the agreement between the two rankings as a check on the surrogate protocol, and prefer the leaderboard whenever it is available for the models being compared.

3.2 Ablation of the Objective

The terms of the objective conflict, and rigidity is the cause. To test whether accuracy and preservation genuinely conflict, we logged the pairwise cosine similarities of all per-term gradients with respect to the shared parameters, along with their norms and shares (Fig. 4). The groups were persistently opposed (cos = -0.25, negative on 80% of steps) at comparable magnitude (ratio of norms about 1.0), excluding weight starvation. Attribution localised nearly all of it to one term: rigidity contributed -0.20 of the -0.25, whereas TLG and the lesion Jacobian contributed about 0.00 despite holding 45% and 5% of their group's norm. TLG was orthogonal to every term (|cos| below 0.05), that is, loud, unopposed and still underperforming, indicating a proxy problem rather than a conflict. Rigidity instead behaved as a regulariser (cos = +0.53 with smoothness, -0.53 with NCC, 0.00 with TLG), so the apparent accuracy-versus-preservation trade-off was, in fact, a regulariser-versus-data-term tension.

FIGURE 4 – Pairwise cosine-similarity matrix of the per-term gradients with respect to the shared parameters, with the per-term gradient-norm shares alongside. Source: pairwise_ cosine_ matrix.png / pairwise_ cosine_ matrix.py in the repository root. This is the most distinctive figure in the paper – no 2025 proceedings paper has an equivalent. Do not cut it.

Fig. 4. Pairwise cosine similarity between the gradients of the individual loss terms. [TODO: caption]

Derivative-based rigidity is ill posed on thin bone. Rigidity opposed bone Dice (-0.45) more than soft tissue Dice (-0.23), which no correct bone penalty should do. Its three conditions are evaluated by finite differences at every voxel and only then masked, so the mask selects which voxels are scored but not which are read. The stencil used for the derivative calculation would therefore also use non-bone voxels. With thin cortical bone and ribs, this dominates. We tested it across five cases: 53.0% (range 51.6 to 54.7) of the voxels entering the loss were not bone; a field with zero displacement inside every bone scored 56 (range 40 to 69), with 100% removed by a single mask erosion; and 44.8% of the gradient magnitude fell outside bone. The term measured the bone-to-soft tissue displacement discontinuity, which any correct registration must exhibit. With the per-structure rigid-fit penalty of Sec. 2.3, the zero displacement field now scores exactly 0 and a per-bone rigid field 2.4e-6, while anisotropic squeeze is still rejected at 1.16.

A surface-distance loss would not help. A per-label analysis of the validation predictions shows that the median and the 75th percentile of the per-label HD95 both equal 2.73 mm, i.e. exactly the in-plane voxel spacing: at least three quarters of all labels already lie at the discretisation floor attainable by a nearest-neighbour–warped label map. The mean HD95 is instead driven almost entirely by a small tail, with the worst 10% of (case, label) pairs contributing 57.6% of the total. This tail is dominated by structures for which no boundary refinement is meaningful: partially visible field-of-view–edge anatomy (brain, skull), limb structures subject to inter-session repositioning (humeri, scapula), and hollow organs with variable filling and hence no well-defined point correspondence (bowel, colon, urinary bladder). A boundary loss would therefore expend its gra-

Table 1. LapIRN variants on the official challenge validation set ($n = 20$ cases), evaluated against the surrogate labels of Sec. 3.1. Each cell reports the mean over cases. DSC and the MTV/TLG errors are given in %, NDV in ppm. Score is the challenge score of Eq. (9); variants are listed in ranking order, highest score first. Bold marks the best value in each column, and the Model and Score entries of the best-scoring variant. * marks metrics on which that variant is significantly better than the variant in the given row (paired Wilcoxon signed-rank, Holm-corrected within each metric, $p < 0.05$).

Model	DSC (%) ↑	HD95 (mm) ↓	MTV %err ↓	TLG %err ↓	NDV (ppm) ↓	Score ↑
V5	88.3	4.77	1.83	1.80	20.7	**69.0**
V6	87.6*	4.93*	1.94	1.64	19.3	63.4
V4	87.5*	5.02*	1.89	**1.64**	6.49	60.3
V9	88.2	4.99	**1.11**	2.14	96.3*	57.7
V2	83.4*	6.08*	2.78	1.79	0.375	43.7
V3	**88.9**	**4.69**	15.0*	16.2*	7.72	33.3
V1	87.2*	5.15*	7.98*	4.99*	156*	29.3
V8	78.8*	6.54*	7.96*	12.2*	7.64	26.1
V7	75.0*	7.00*	2.93	5.98*	**0.00**	22.8

dient on labels that are already optimal, while leaving the tail — a capture-range and correspondence problem rather than a boundary-precision one — unaffected.

3.3 Comparison of LapIRN Variants

Table 1 compares the nine LapIRN variants. V5 attains the highest challenge score (69.0), with 88.3% DSC, 4.77 mm HD95 and MTV and TLG errors of 1.83% and 1.80%. Its registration accuracy is significantly better than that of every other variant except V3, which reaches a marginally higher DSC (88.9%) but degrades the PET biomarkers by an order of magnitude (15.0% MTV and 16.2% TLG error), illustrating that accuracy and biomarker preservation trade off against each other. Six variants produce fewer non-diffeomorphic voxels than V5, but all of them lie below the 50 ppm practical-equivalence threshold, so the difference does not affect the ranking.

3.4 Comparison against Baselines

Table 2 places the selected variant against the baselines. It improves DSC from 73.3% (NiftyReg, the strongest baseline) to 88.3% and halves HD95 from 7.09 to 4.77 mm, while reducing the MTV and TLG errors by more than a factor of two and the non-diffeomorphic volume from 1525 to 21 ppm. All differences are significant on every metric (paired Wilcoxon signed-rank, Holm-corrected, $p < 0.01$).

Table 2. The selected variant against the baselines on the official challenge validation set ($n = 20$ cases), evaluated against the surrogate labels of Sec. 3.1. Each cell reports the mean over cases. DSC and the MTV/TLG errors are given in %, NDV in ppm. Bold marks the best value in each column. * marks metrics on which our method is significantly better than the baseline in the given row (paired Wilcoxon signed-rank, Holm-corrected within each metric, $p < 0.05$). Dashes mark metrics that are undefined for a baseline: the unregistered and affine cases have no deformation field, and the PET biomarkers are only meaningful after resampling.

Method	DSC (%) ↑	HD95 (mm) ↓	MTV %err ↓	TLG %err ↓	NDV (ppm) ↓
Initial	14.8^*	30.9^*	–	–	–
Affine	52.9^*	11.7^*	5.47^*	4.70^*	–
NiftyReg	73.3^*	7.09^*	4.46^*	8.68^*	1525^*
ConvexAdam	63.0^*	8.41^*	5.10^*	4.15^*	100^*
Ours	**88.3**	**4.77**	**1.83**	**1.80**	**20.7**

3.5 Validation Leaderboard

3.6 Qualitative Results

4 Discussion

[TODO: write the discussion following the outline above]

[TODO: optional short conclusion]

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Disclosure of Interests. It is now necessary to declare any competing interests or to specifically state that the authors have no competing interests. Please place the statement with a bold run-in heading in small font size beneath the (optional) acknowledgments[4], for example: The authors have no competing interests to declare that are relevant to the content of this article. Or: Author A has received research grants from Company W. Author B has received a speaker honorarium from Company X and owns stock in Company Y. Author C is a member of committee Z.

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[4] If EquinOCS, our proceedings submission system, is used, then the disclaimer can be provided directly in the system.

FIGURE 5 – Standard qualitative panel: fixed, moving, warped and difference image, plus a $\det J_\varphi$ *heat map and a zoom on an intraosseous lesion showing that the lesion volume is preserved. Include one failure case (bowel or bladder filling) – the 2025 papers that showed a failure case read as more honest.*

Fig. 5. Qualitative registration results. [TODO: caption]

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